

MNIST Dataset Analysis and Exploratory Data Analysis

Dataset Overview and Basic Statistics: The MNIST handwritten digit dataset contains **10,000 test samples** with **784 features** (28×28 pixel images) representing digits 0-9. Each pixel value ranges from **0 to 255**, where 0 represents black and 255 represents white.

label	1x1	1x2	1x3	1x4	1x5	1x6	1x7	1x8	1x9	...	28x19	28x20	28x21	28x22	28x23	28x24	28x25	28x26	28x27	28x28
0	7	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0
1	2	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0
2	1	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0
3	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0
4	4	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0

5 rows × 785 columns

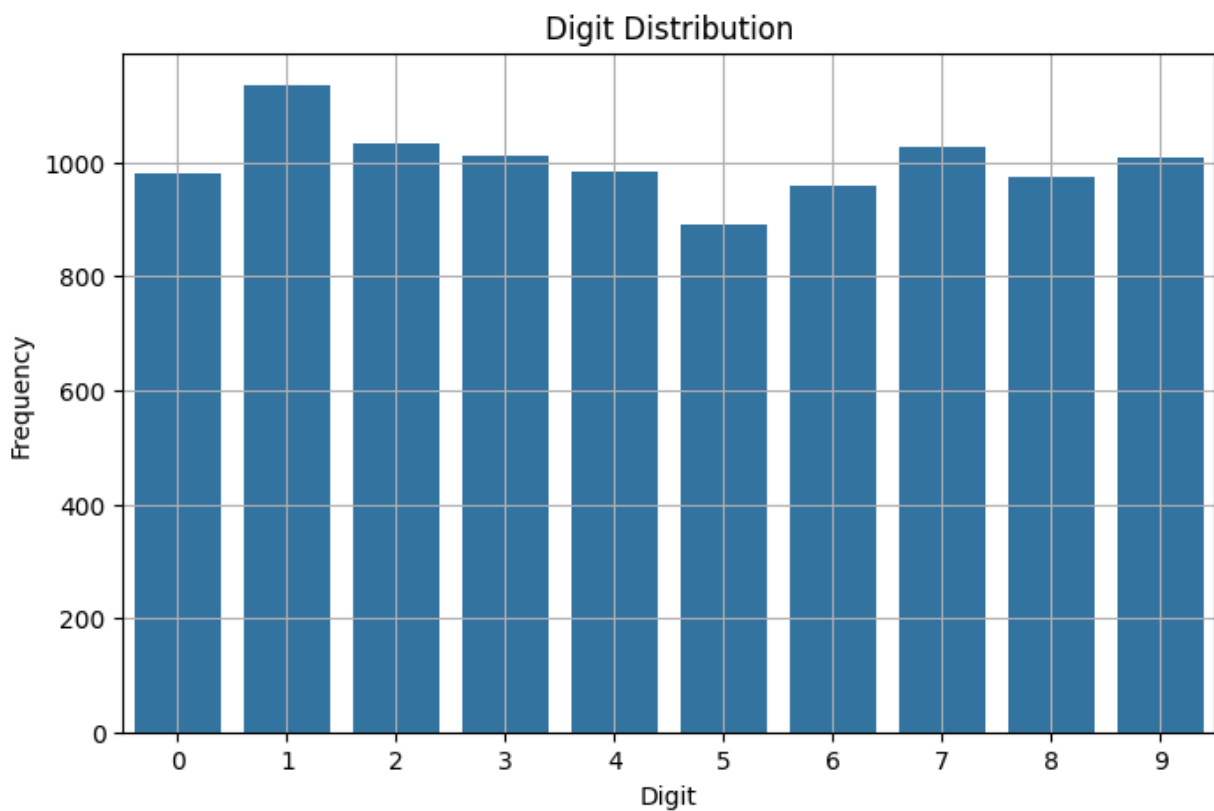
Statistical Summary:

- **Mean pixel value:** 33.31
- **Median pixel value:** 0.0
- **Mode pixel value:** 0 (most pixels are background)
- **Minimum value:** 0 (pure black)
- **Maximum value:** 255 (pure white)
- **Data type:** Integer values
- **Missing values:** 0 (complete dataset)

```
Mean: 33.791224489795916
Median: 0.0
Mode: 0
Name: 0, dtype: int64
Min: 0
Max: 255
```

Data Distribution and Quality Assessment:

Class Distribution Analysis : The dataset shows a balanced distribution across all 10 digit classes, with each digit appearing approximately 1,000 times (10% each). This balanced nature ensures unbiased model training.



Data Quality:

- **Complete dataset:** No missing values detected
- **Duplicate images:** Minimal duplicates found in sample analysis
- **Black images:** 0 completely black images
- **Sparse images:** Few images with fewer than 50 active pixels

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=== DATA QUALITY ANALYSIS ===

Dataset shape: (10000, 784)
Number of features: 784 (28x28 pixels)
Number of samples: 10000
Data type: int64
Missing values: 0
Pixel value range: 0 to 255

Class distribution:
Digit 0: 980 samples (9.8%)
Digit 1: 1135 samples (11.3%)
Digit 2: 1032 samples (10.3%)
Digit 3: 1010 samples (10.1%)
Digit 4: 982 samples (9.8%)
Digit 5: 892 samples (8.9%)
Digit 6: 958 samples (9.6%)
Digit 7: 1028 samples (10.3%)
Digit 8: 974 samples (9.7%)
Digit 9: 1009 samples (10.1%)

Checking for duplicates in sample of 1000 images...
/tmp/ipython-input-368133002.py:163: FutureWarning: Series.__getitem__ treating keys as positions is deprecated. In a future version, integer keys will always be treated as labels (consistent with DataFrame
print(f'Data type: {X.dtypes[0]}")
Duplicate images in sample: 0

Image property analysis:
Completely black images: 0
Very sparse images (< 50 pixels): 10

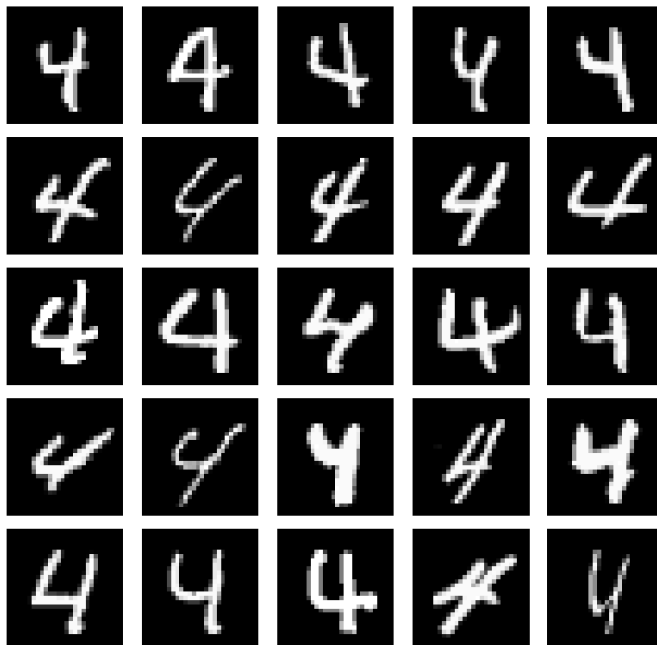
```

Visual Pattern Analysis and Image Characteristics

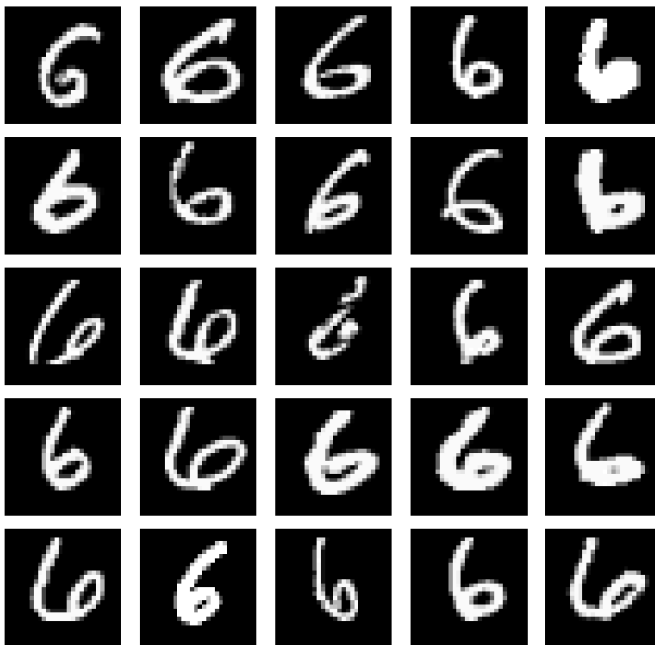
Sample Digit Variations: Analysis of 25 random samples per digit class reveals significant **handwriting variations**. Key observations:

- **Digit 4:** Shows confusion with digit 9 in some cases
- **Digit 6:** Occasionally similar to digit 0
- **Digit 7:** Some samples resemble digit 1
- **Digit 8:** Most consistent across samples

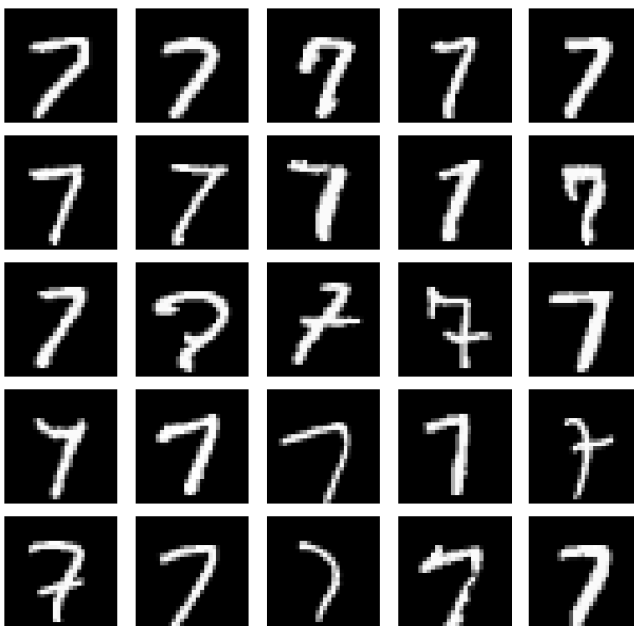
Sample Images of Digit 4



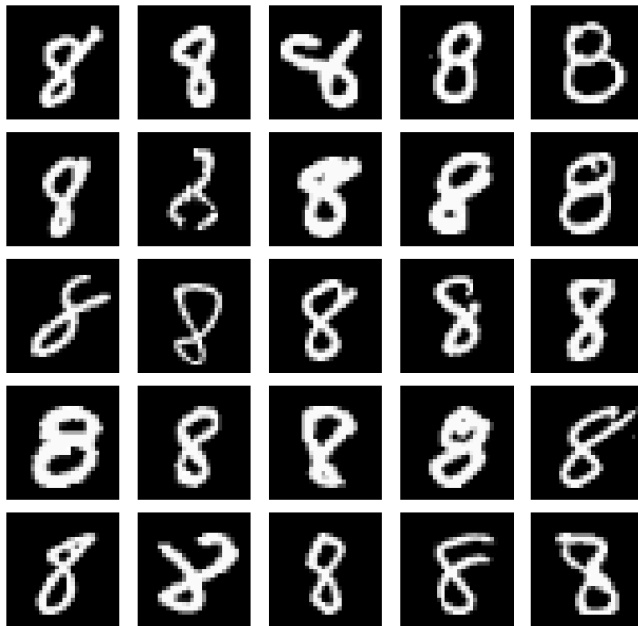
Sample Images of Digit 6



Sample Images of Digit 7



Sample Images of Digit 8

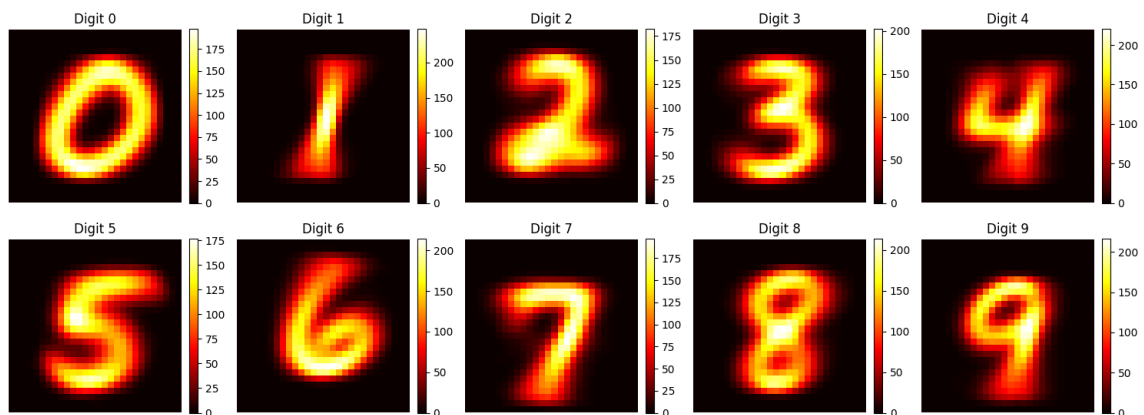


Average Digit Templates (Heatmaps)

The mean pixel intensity heatmaps reveal **common stroke patterns** for each digit:

- **Bright regions:** Indicate consistent pen strokes across all samples
- **Dark regions:** Show areas rarely touched by writing
- **Central concentration:** Most informative pixels located in image center

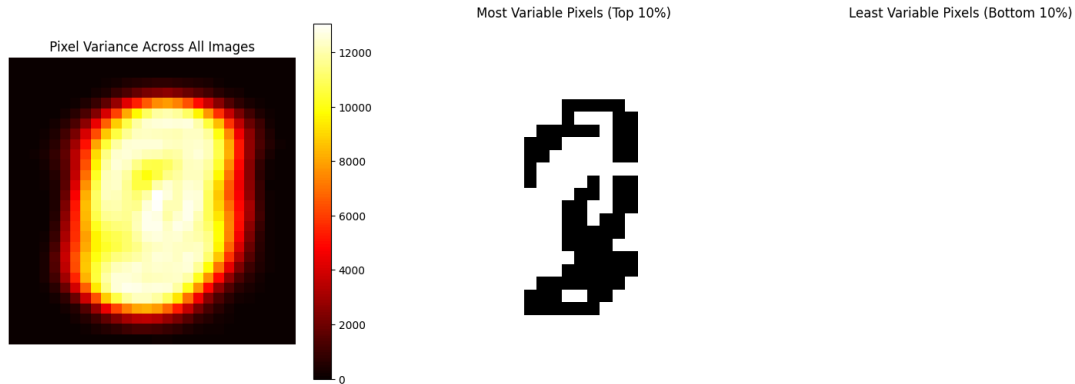
Average Digit Templates (Heatmaps)



Pixel Importance Analysis

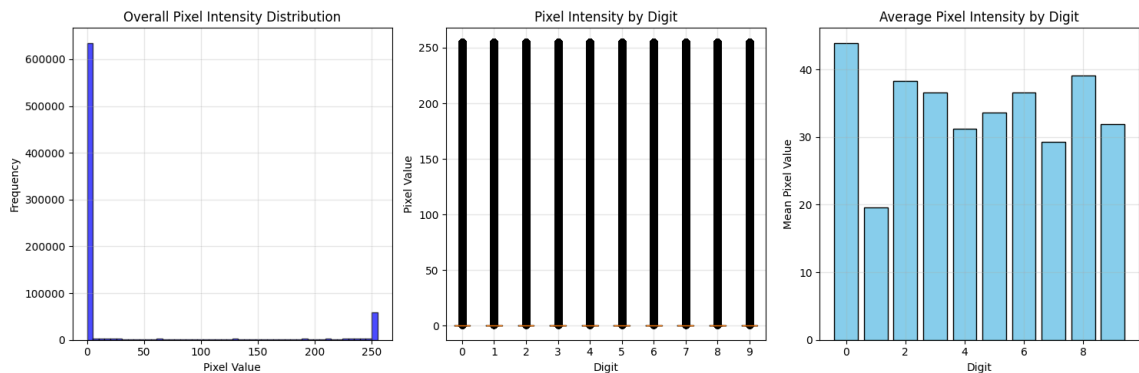
Feature analysis identifies **most and least informative pixels**:

- **High-variance pixels:** Located in central 14×14 region where digits are typically drawn
- **Low-variance pixels:** Found in border regions (typically background)
- **Edge pixels:** Contribute minimal information for digit classification



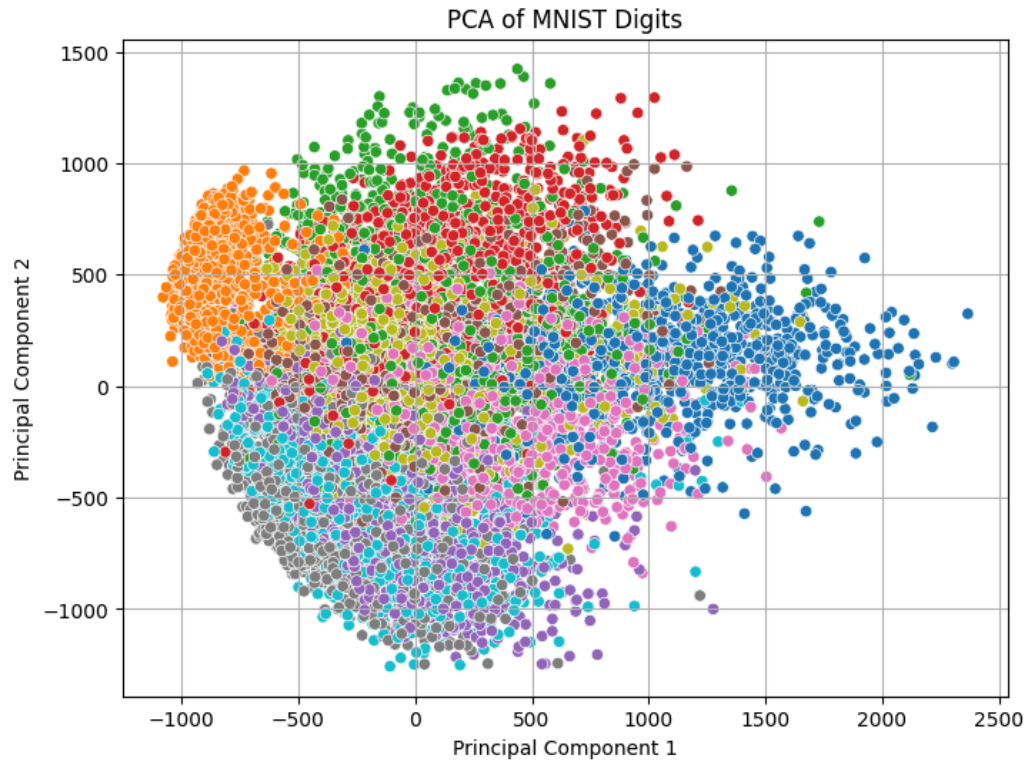
Multi-faceted pixel analysis reveals:

- **Bimodal distribution:** Most pixels are either 0 (background) or high values (foreground)
- **Digit-specific patterns:** Different digits show varying mean pixel intensities
- **Box plot analysis:** Digit 1 shows lowest variance, digit 8 shows highest complexity



Dimensionality Reduction Insights & PCA Analysis (2D Projection)

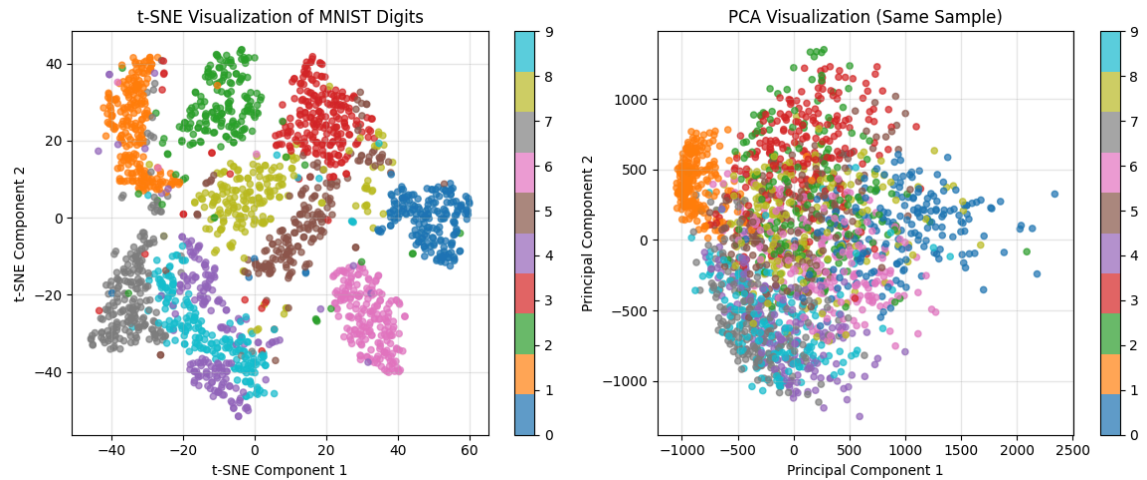
Principal Component Analysis reveals **overlapping clusters** in 2D space, indicating that linear dimensionality reduction provides limited class separation for this complex dataset.



t-SNE Visualization (Advanced Clustering)

t-SNE provides **superior cluster separation** compared to PCA, showing:

- **Distinct digit clusters:** Better separation of digit classes
- **Outlier identification:** Unusual handwriting samples at cluster boundaries
- **Similar digit proximity:** Digits 6-8-9 and 3-5-8 show expected proximity due to visual similarity



Data Quality Assessment:

High-quality dataset: No missing values, minimal noise, well-balanced classes. **Appropriate resolution:** 28×28 pixels is sufficient for digit recognition. **Consistent format:** Standardized preprocessing evident

Pattern Recognition Insights:

- **Class separability:** Some digits (1, 0) more distinct than others (4, 9)
- **Feature importance:** Central pixels most informative for classification
- **Complexity variation:** Digit 8 most complex, digit 1 most consistent

Recommendations for Model Development:

1. **Preprocessing:** Minimal needed due to clean dataset
2. **Feature selection:** Focus on central pixel regions
3. **Class handling:** Pay attention to 4-9 and 3-8 confusion pairs
4. **Augmentation:** Consider rotation/scaling for improved robustness

Tools Used: Python with pandas, matplotlib, seaborn, scikit-learn **Dataset Source:** Kaggle MNIST CSV format (oddrationale/mnist-in-csv) **Analysis Scope:** Complete exploratory data analysis following KDNuggets 7-step methodology **Visualizations:** Multiple graph types from Python Graph Gallery including heat